

C A S E S E R I E S

Intense pulsed light in treatment of pigmentary disorders in Indian skin- A Case Series

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ABSTRACT

Introduction: Intense pulsed light (IPL) treatment has become very popular in India. However, there is limited literature regarding its use for pigmentary lesions in Indian skin.

Objectives: To evaluate the safety and effectiveness of IPL in the treatment of pigmentary disorders in Indian patients.

Methods: IPL treatment was offered to 11 Indian patients with pigmentary disorders.

Results: Good outcomes were observed in patients with lentigines, freckles, and post-acne pigmentation. No serious adverse effects were reported.

Conclusions: IPL appears to be a safe and effective treatment option for selected pigmentary disorders in Indian skin.

Key words: intense pulsed light, lentigenes, post-acne pigmentation



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Introduction

Intense pulse light was first developed in 1992 by Doctors Goldman, Fitzpatrick and Eckhouse for the treatment of leg telangiectasias and the first FDA approved IPL device was marketed in 1995¹. IPL devices are unique in that they use flashlamps and bandpass filters to produce pulsed light of varying wavelengths, durations, and fluences. As a result, IPL devices can be used to treat a wide range of conditions. Polychromatic light of a broad wavelength spectrum of approximately 400-1400 nm is emitted, which is absorbed by different chromophores like water, haemoglobin and melanin. They get heated up and are destroyed by thermocoagulation. The United States Food and Drug Administration (FDA) has approved IPL for telangiectasias, photo rejuvenation, facial wrinkles, hyperpigmentation, lentigines, ephelides, melasma, rosacea, acne vulgaris, poikiloderma of Civatte, port-wine stains, haemangiomas, leg veins, venous malformations, and removal of unwanted hair. The pulse duration of IPL systems can be set to ranges between 0.2 and 100 milliseconds and is usually kept lower than the thermal relaxation time of the target structure to protect the surrounding tissue². The fact that IPL devices emit light with pulse durations in the millisecond range makes them less effective than Q-switched lasers for the treatment of pigmented lesions³. Lasers usually used for pigmentary lesions are the Q-switched 532nm Nd: YAG laser & the Q-switched ruby laser. Q-switched lasers deliver energy with high fluence and very short pulse duration in nanoseconds (5-100ns), which is much shorter than the thermal relaxation time of melanosomes. This causes fragmentation of melanosomes and pigment granules by photomechanical effect. This does not happen with the IPL. Picosecond and nanosecond lasers with a wavelength of 1064 nm and 755 nm are also efficient in treating pigmented lesions. Some studies have shown the utility of IPL in pigmented lesions in Asians^{4,5}. However, there are no reports in the literature on Indian patients with pigmentary disorders.

Report

A total of 11 patients were treated for pigmentary disorders in our centre over a period of 6

months. Approval from the Institutional Ethical Committee, Government General Hospital, Chennai (K. Dis. No 25406/P & D3/Ethics/Dean/GGH/07) was obtained. Among the patients, one patient (Becker's nevus) dropped out of the study after a single sitting. An average of 3.1 treatment sessions was needed in the treatment of pigmentary disorders. The IPL i 200 system (Cyden Ltd, Swansea, UK) was used for treatment, employing a spot size of 33 x 27 mm and having a wavelength spectrum of 530nm-1200nm. Treatment parameters were program 15(triple sub-pulses with total pulse durations in the range of 35-45 ms with intra-pulse off-times of 5-10 ms to facilitate epidermal cooling) and fluence of 10 – 15 J/cm². The clinical endpoint was darkening of treated area. Digital photographs were taken before each treatment session and based on the photographs, improvement was graded by independent dermatologists as poor (<25% improvement), moderate (25-50% improvement), good (50-75% improvement) and excellent (75% improvement). Patients rated their satisfaction using a three-point scale (not satisfied, satisfied and very satisfied) at the end of sessions. Treatment sessions were repeated after 3 weeks. Patients were advised emollients for 2 days and to resume topical bleaching agents (2% azaleic acid cream) after 2 days. Good results were seen in lentigines (Figure 1) and freckles. Good results were seen in post-acne pigmentation (Figure 2) while moderate improvement was seen in a case of melasma. Poor improvement was seen in Nevus of Ota (Figure 3) and the other case of melasma. Limited improvement was observed in melanocytic nevi and post-inflammatory hyperpigmentation. Overall, 27% of patients had good results and 54% of patients were satisfied with the procedure after a mean of 3.1 treatment sessions (Table 1).

Discussion

Unlike lasers, the light emitted by IPL systems is polychromatic, noncoherent, and noncollimated (420–1,400 nm) with varying pulse durations. Consequently, IPL is generally less selective than laser systems⁶. The mechanism of action of IPL for treating pigmentary lesions is thought to be due to rapid differentiation



Figure 1. Good improvement seen in lentigines after 3 sessions.



Figure 2. Good improvement seen in post-acne pigmentation after 2 sessions.



Figure 3. Poor improvement seen in Nevus of Ota after 4 sessions.

of keratinocytes induced by thermal heating. This process causes an upward migration of melanosomes along with necrotic keratinocytes, resulting in their

shedding as the microcrusts are removed from the skin surface.³ IPL devices have been used to treat benign pigmented lesions including ephelides and

Table 1. Details of patients with pigmentary disorders who received IPL therapy.

S.No	Age	Sex	Indication	Skin type	No of sittings	Improvement	Satisfaction	Adverse effects
1	27	M	Post-inflammatory hyperpigmentation	3	3	Poor	Not satisfied	Nil
2	30	F	Melasma	3	2	Moderate	Satisfied	Transient erythema
3	40	F	Lentigenes	3	3	Good	Satisfied	Nil
4	27	M	Freckles	4	4	Good	Satisfied	Nil
5	29	F	Melasma	3	4	Poor	Satisfied	Nil
6	22	F	Nevus of ota	4	4	Poor	Satisfied	Nil
7	28	M	Post-acne hyperpigmentation	5	3	Poor	Not satisfied	Nil
8	45	F	Post-inflammatory pigmentation	4	4	Poor	Not satisfied	Nil
9	30	M	Melanocytic nevus	5	2	Poor	Not satisfied	Transient pain
10	30	F	Post-acne pigmentation	3	2	Good	Very satisfied	Nil
11	15	F	Becker's Nevus	4	Drop out	-	-	-

solar lentigenes, with improvement usually seen after a few monthly treatment sessions⁶. In our series, an average of 3.1 treatment sessions was carried out for pigmentary disorders after which treatment response was assessed. Of the two patients with melasma, one had moderate improvement (25-50%), and the other showed poor improvement, photographs being assessed by two independent dermatologists. Patients were asked to continue a regimen of topical bleaching agents and sunscreens and continue it throughout the course of the treatment. These patients had been on topical bleaching agents and sunscreens for some time without any change. Any subsequent improvement was attributed to IPL treatment. Hence, as in acne, the cost effectiveness of the whole treatment must be kept in mind while considering IPL. One case of post-acne hyper pigmentation had good response similar to that obtained by Pacquet and Pierard who got 80% improvement in facial post-inflammatory hyper pigmentation⁷. We did not get satisfactory results in melanocytic nevi though Bjerring et al⁸ have reported 66.3% improvement in melanocytic nevi. This may be related to the darker skin phototypes of our patients or because we only tried it in a single patient. Wang et al compared the efficacy of Q-Switched Alexandrite

Laser (QSAL) and IPL in the treatment of freckles and found the QSAL superior; but post-inflammatory hyper pigmentation was seen only in QSAL, suggesting that IPL may represent a safer option for Asian patients⁹. A recent meta-analysis on IPL in melasma found it to be efficacious, with very few incidents of adverse effects¹⁰. A retrospective analysis found that IPL does not have long term benefits in Becker's nevus¹¹. Li et al found good response to IPL in post-burn hyperpigmentation in Chinese Han patients¹². They observed that response to IPL was best when treatment started early, i.e within 6 months. We did not get good results in post-inflammatory hyperpigmentation.

Our data is limited by the small number of patients. Response to IPL was varied in different patients. The indications for treating pigmentary disorders in our study were very varied. Each condition may need different number of sessions and device settings. Further studies concentrating on a single pigmentary disorder in Indian patients should be planned and are the way forward. In conclusion, IPL appears to be a safe treatment modality for selected pigmentary lesions in Indian patients, but cost effectiveness must be considered. Ideally, treatment settings should be conservative to prevent unwanted side effects.

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